

Effect of Neutronic activation in the Biodegradable Chitosane polymer associated with Gold Nanoparticles.



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1. INTRODUCTION

Recent studies investigate the effects of ionizing radiation on organic molecules for medical applications, which biopolymers are widely studied due to their biological properties and structural stability. Chitosan, a biopolymer derived from chitin, has been widely studied in the biomedical field due to its biological properties, including antibacterial and anticancer activities, as well as its role as a drug carrier. When combined with gold nanoparticles (AuNPs), which exhibit favorable physicochemical properties and high applicability in medicine, a promising nanosystem is formed, particularly for use in cancer diagnosis and treatment. This study aims to develop this nanosystem and assess the effects of neutron activation on the chitosan structure, focusing on whether any degradation affects its functionality.

2. MATERIALS AND METHODS

AuNPs were synthesized via the Turkevich reduction method and incorporated into chitosan during polymer preparation following Hosseini method (Figure 1). The samples (Figure 2) were irradiated in the J9 channel of the Argonauta reactor with a neutron flux of $10^8 \text{ n.cm}^{-2} \cdot \text{s}^{-1}$ for 2 hours and characterized using measurements of activity by dose calibrator, XRF, and gamma spectrometry to evaluate structural integrity and detect possible interfering radionuclides.

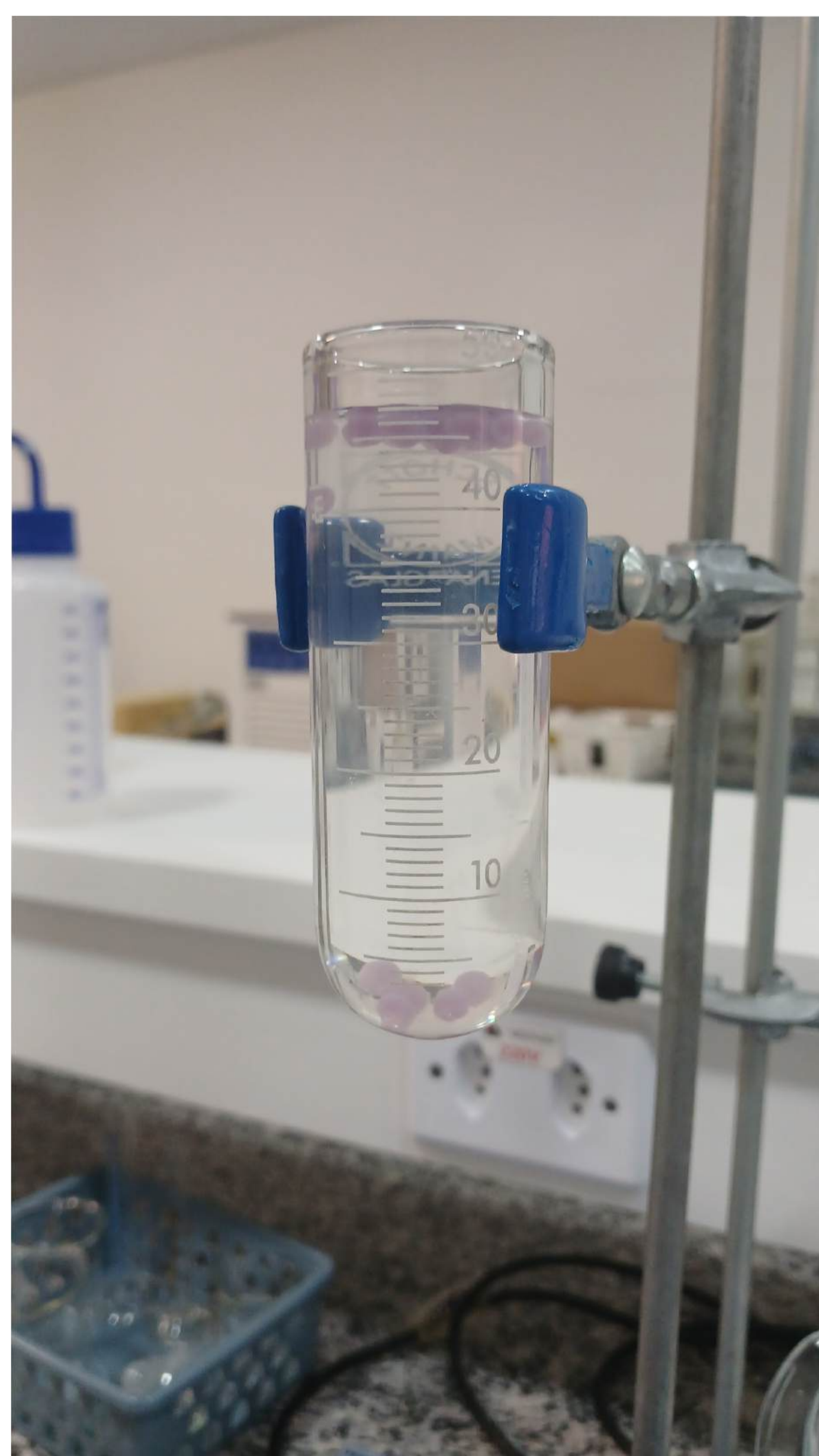


Figure 1: Drip-casting process of Au-CHI's microparticles.

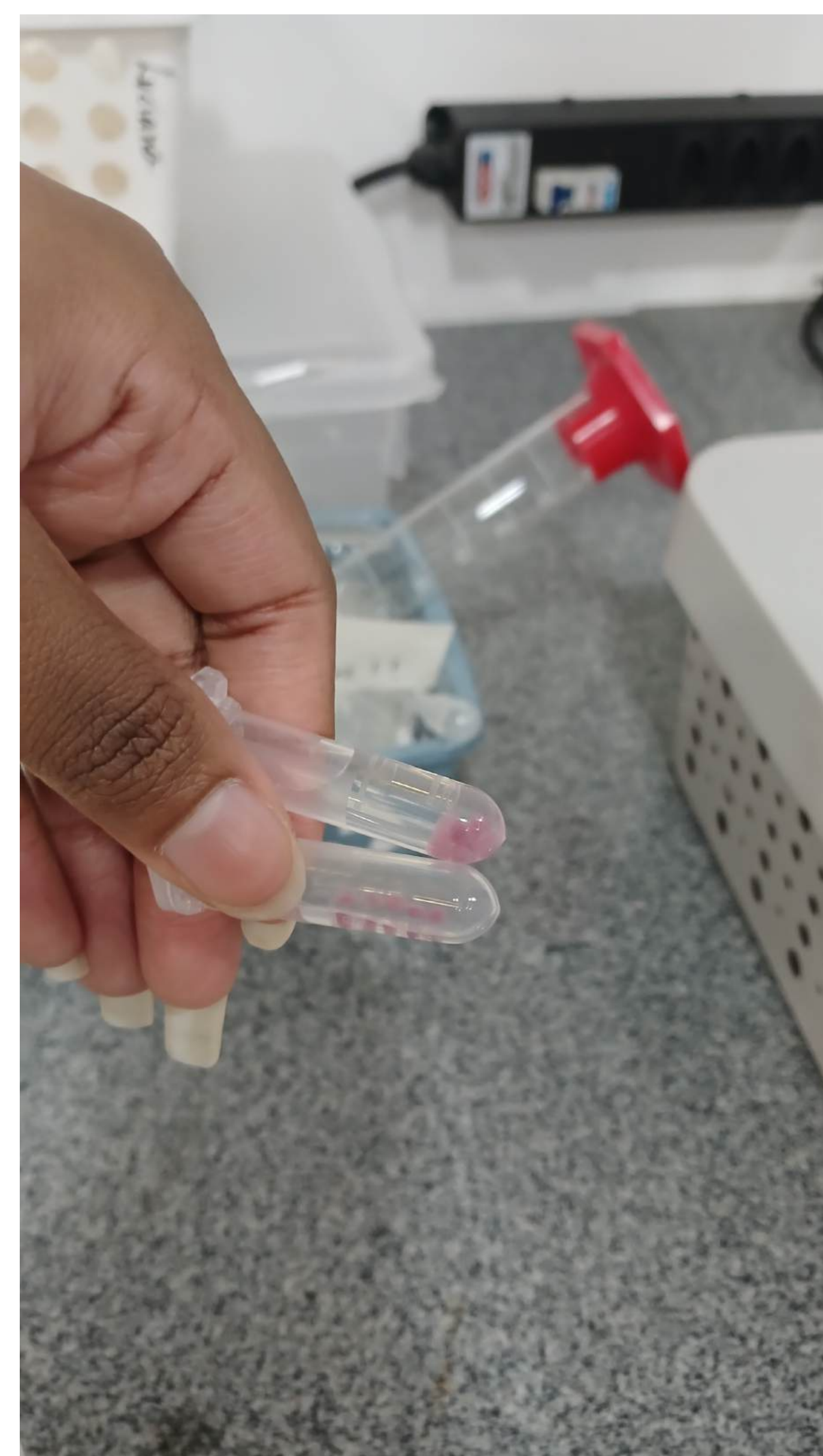


Figure 2: Au-CHI's samples.

3. RESULTS

ACTIVITY

Table 1: Activity measurement of the Au-CHI system.

Sample	Description	Activity (MBq)
1	Au-CHI	0,083
2	STPP	0,026

XRF

Table 2: Results of XRF Analyses.

Analyte	Concentration (%)	Standard Deviation (%)
P	88.936	6.541
Au	5.946	1.881
Cu	3.023	0.868
Fe	2.095	1.227

GAMMA SPECTROMETRY

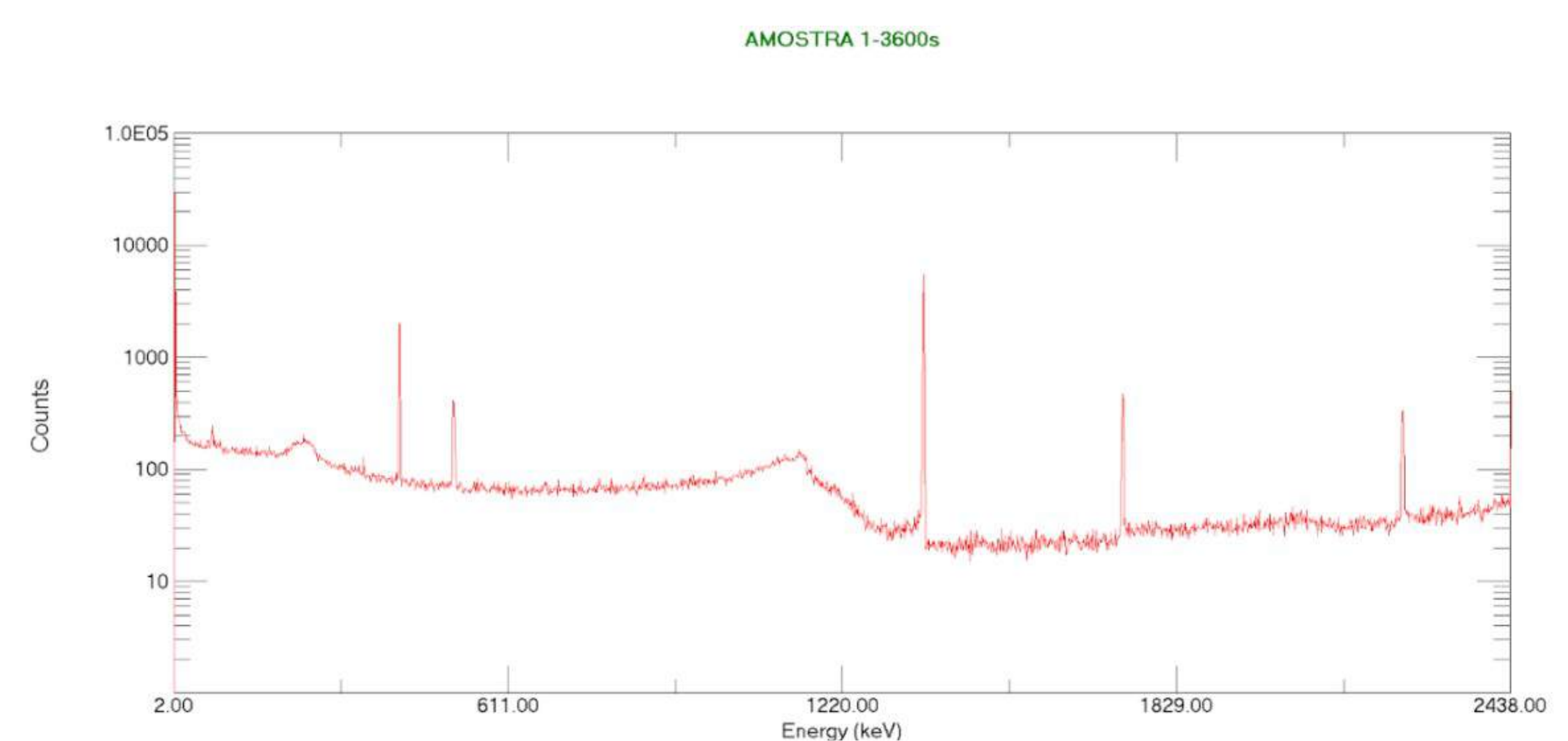


Figure 3: Spectrum of the Au-CHI's system obtained by Gamma Spectrometry.

4. CONCLUSIONS

This work demonstrates the potential of gold nanoparticle associated with chitosan (Au-CHI) nanoplatform as a promising theranostic agent in nuclear medicine. Ensuring its structural stability under neutron irradiation is essential for safe and effective applications. Future directions point toward optimizing this system for personalized radiopharmaceuticals, enabling precise cancer treatment with reduced damage to healthy tissues.